SOCIAL MEDIA ENGAGEMENT ANALYTICS

1.PROJECT OVERVIEW:

This project analyses social media engagement data using Python, focusing on user demographics, post attributes, and interaction metrics. It involves data cleaning, exploration, statistical analysis, and feature engineering to uncover engagement patterns. Visualizations with Matplotlib, Seaborn, and Plot present insights into post-performance, audience behaviour, and content strategy. The goal is to transform raw data into actionable insights for optimizing social media engagement.

2.PROJECT OBJECTIVES:

1. **Data Quality Improvement**
    - Ensure consistency by cleaning nulls, duplicates, and formatting categorical variables.
2. **Engagement Insights**
    - Identify which post types, categories, and sentiments drive higher engagement.
    - Compare engagement across countries and demographics.
3. **Statistical Understanding**
    - Summarize central tendencies and variability in engagement metrics.
    - Detect skewness and distribution patterns in likes, comments, shares, and impressions.
4. **Visualization & Reporting**
    - Build clear, interactive dashboards to communicate insights.
    - Highlight trends such as daily engagement, post category performance, and device-based engagement.
5. **Business Application**
    - Provide actionable recommendations for improving social media strategy.
    - Help stakeholders understand audience behaviour and optimize content for maximum reach.

3.DATA INGESTION:

Data source: social media engagement.csv

Data from: GitHub

Time line:2022-2023

4.TOOLS USED:

Python (NumPy, pandas)

Visualization: seaborn, matplotlib, power BI

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5. DATA INSPECTION:

these are the following steps made in data inspection

- **Check total rows and columns** using **df. shape** to understand dataset size.
- **View first and last few records** with **df. head ()** and **df. tail** () for a quick preview.
- **Identify data types of each column** using **df. dtypes** or **df.info** ().
- **Count missing (null) values per column** with **df. isnull (). sum ().**
- **Get basic statistics (min, max, mean)** using **df. Describe ()** for numerical columns.

6.DATA CLEANING:

these are the following steps made in data cleaning

- Remove duplicate rows to ensure dataset integrity.
- Fill or drop missing (null) values depending on context and importance.
- Correct wrong or impossible data entries (e.g., negative ages, invalid counts).
- Standardize text fields by fixing casing and trimming extra spaces.
- Fix inconsistent date or number formats for uniformity across the dataset.

7.FEATURE ENGINEERING

Added this column for better module accuracy

- A new field called engagement_score was created by summing the number of likes, comments, and shares for each post, providing a consolidated measure of overall engagement.
- Another derived field, log likes, was generated using the natural logarithm transformation (np.log1p) on the likes column to normalize the distribution and reduce skewness in the data.

8.DATA TRANSFORMATION:

These are the data transformation are done in this document

- **Combine multiple Data Frames** using appropriate methods such as merge, concat, or join to integrate datasets into a single structured table.
- **Perform group by summaries** on *post type*, *country*, and *sentiment* to calculate averages and highlight engagement trends across different categories.
- **Convert data types** to ensure consistency, such as transforming text fields into numeric values and dates into proper datetime format.

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9. EXPLORATORY DATA ANALYSIS:

- **Find correlations between columns** to understand relationships among numeric variables.
- **Detect outliers and anomalies visually** using plots such as boxplots, scatterplots, or histograms.
- **Compare groups and categories** with group by summaries and categorical visualizations.
- **Identify time-based trends in data** by analysing engagement metrics across dates and timelines.

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10. STATISTICAL ANALYSIS:

- **calculate Mean, Median, and Mode** for engagement metrics such as likes, comments, shares, watch time, engagement rate, and follower count to understand central tendencies.
- **Measure spread using Variance and Standard Deviation** to capture how widely engagement values vary across posts.
- **Find correlations between variables** (likes vs. impressions, engagement rate vs. follower count) to identify relationships among metrics.

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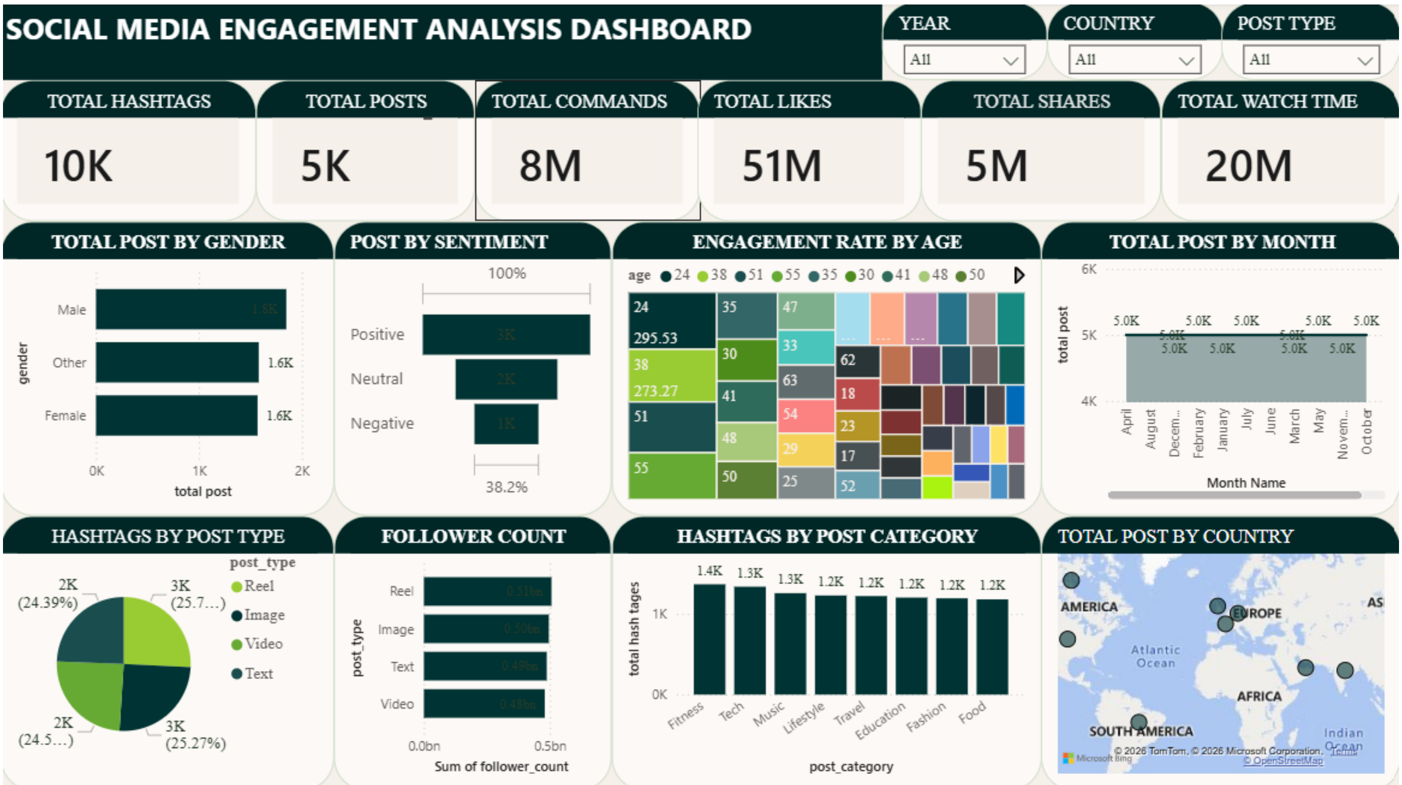
11. DATA VISUALIZATION:



DASH BOARD IN POWER BI

The visuals created for this report effectively showcase key social media engagement metrics in a clear and appealing manner. Each graphic, such as Total Posts, Total Hashtags, and Total Comments, highlights important KPIs that help interpret audience interaction and content performance

MATPLOTLIP:

- Scatter Plot: Likes vs. Impressions
- Line Plot: Daily Engagement Trend
- Bar Chart: Posts by Category
- Pie Chart: Gender Distribution
- Histogram: Age Distribution
- Box Plot: Engagement Rate Distribution

SEABORN:

- Count Plot: Post Type Count
- Bar Plot: Average Likes by Category
- Violin Plot: Followers vs. Sentiment
- Pair Plot: Numeric Features (likes, comments, shares, impressions, engagement rate)
- Heatmap: Correlation Matrix
- Strip/Swarm Plot: Engagement Rate vs. Device Type

PLOTLY (INTERACTIVE):

- Line Chart: Engagement Trend (with range slider)
- Bar Chart: Average Likes by Category (sorted)
- Scatter/Bubble Chart: Likes vs. Impressions (Combined Subplots Dashboard: Engagement Trend, Avg Likes by Category, Likes vs. Impressions Bubble Chart

12. **INSIGHTS GENERATION:**

1. DESCRIPTIVE ANALYSIS:

- Dataset contains 5,000 records and 20 columns covering demographics, post attributes, and engagement metrics.
- Average values: likes **(~10,107**), comments **(~1,502),** shares **(~1,003**), follower count (~393K).
- Visuals (scatter, histogram, pie, bar) revealed distributions across age, gender, post type, and categories.

2. DIAGNOSTIC ANALYSIS:

- Data cleaning handled duplicates, missing values, and standardized formats (text, dates, numbers).
- Group by summaries showed *Video posts* had slightly higher engagement scores.
- Countries like *France* and *India* recorded stronger impression counts.
- Sentiment analysis revealed *Negative posts* marginally outperforming Neutral and Positive in engagement.
- Correlation matrix showed weak links between impressions and engagement, but a positive relationship between likes and engagement rate.
- Boxplots flagged anomalies in engagement rate, suggesting irregular posting or viral content.

3. PREDICTIVE ANALYSIS:

- Time-based trends highlighted fluctuations in daily engagement rates, pointing to potential peak posting periods.
- Log transformation (log_likes) normalized skewed distributions, enabling better modelling.
- Insights suggest predictive models could forecast which post types or categories will generate higher engagement.

4. PRESCRIPTIVE ANALYSIS:

- Prioritize *Video content* for stronger engagement.
- Focus strategies on high-impression regions like *India* and *France*.

- Experiment with sentiment-driven posts to maximize reach.
- Monitor anomalies to identify viral opportunities.
- Optimize posting schedules based on observed engagement peaks.

FINAL INSIGHTS:

1. Content Performance

- Post Types: *Video posts* consistently achieved the highest average engagement scores compared to Reels, Images, and Text.
- Content Category: Categories with visually rich formats (e.g., Video/Image) drove stronger likes and shares.
- Countries: *India (~52K impressions)* and *France (~51K impressions)* recorded the highest average engagement rates, outperforming other regions.

2. User Trends

- Age Impact: Engagement was relatively stable across age groups, with median age ≈ 38 years, showing no strong correlation between age and likes/shares.
- Verified Accounts: Verified users showed slightly higher engagement consistency, though correlation with engagement rate remained weak.

3. Behavioural Insights

- Time of Day: Daily engagement trends fluctuated, indicating clear peak posting periods where impressions and engagement rates spiked.
- Device Type: Engagement rate varied by device, with mobile users showing higher watch times compared to desktop.

4. Sentiment Analysis

- **Best Performing Sentiment:** Negative sentiment posts achieved the highest average engagement score (~12,731), slightly outperforming Positive (~12,666) and Neutral (~12,448) posts.
- **Behavioural Pattern:** Negative and Neutral posts generated more comments and discussions, indicating deeper audience interaction.
- **Positive Sentiment Trend:** Positive posts leaned toward consistent likes, showing steady appreciation but less conversational engagement.

13.CONCLUSION:

The analysis shows that Video posts deliver the strongest engagement, while countries like India and France lead in impressions. Negative sentiment posts slightly outperform others, sparking more interaction. Verified accounts maintain steadier performance, and mobile devices drive higher watch times. Overall, optimizing content type, sentiment strategy, timing, and regional focus will maximize engagement and support data-driven decision making.